COL774/COL7341

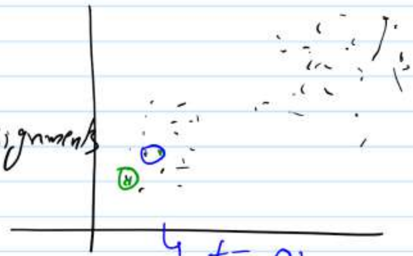
Nov 4, 2025

Last Class:

$\{x^{(i)}\}_{i=1}^{m}$ K: # of clusters

Objective:- $\{c^{(i)}\}_{i=1}^{m}$:- cluster Assignments

$c^{(i)} \in \{1 .. K\}$



$\hookrightarrow t=0;$

$\{\mu_k\}_{k=1}^{K}$ $\mu_k \in \mathbb{R}^n$

$$J(c,\mu) = \frac{1}{m}\sum_{i=1}^{m} \|x^{(i)} - \mu_{c^{(i)}}\|^2$$

1st Iteration -

(E-Step) (M-Step)

$\Rightarrow$ Iterative Algorithm - K-Means Algorithm

$t \leftarrow 0;$

$\mu^{(t)} \leftarrow \text{init}();$

do{

E-step $c^{(t+1)} \leftarrow \arg\min_c J(c, \mu^{(t)});$

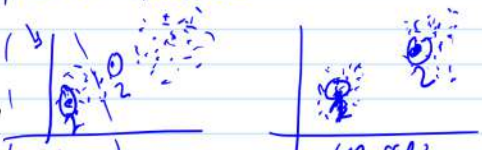
M-step $\mu^{(t+1)} \leftarrow \arg\min_\mu J(c^{(t+1)}, \mu);$

$t \leftarrow t+1;$

} (while ! converged)

convergence:- $c^{(t)} = c^{(t+1)}$

IInd Iteration -



(E-step) (M-Step)

IIIrd iteration

(E-Step) (M-Step)

$\hookrightarrow$ IVth iteration $\to$ converged

K-Means will "converge" to "local minima" of $J(c,\mu)$ (Assuming it converges)

$$J(c,\mu) :- \frac{1}{m}\sum_{i=1}^{m}\|x^{(i)} - \mu_{c^{(i)}}\|^2$$

K-Means algorithm can be seen as doing Block Coordinate Descent over the above objective function over (c, μ)

$x^{(i)} \in \mathbb{R}^n$

K:- # of clusters

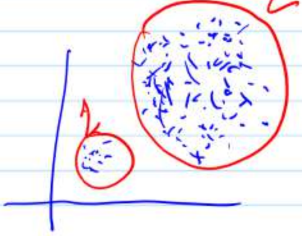
m:- # of points

Time complexity :- T $\hookrightarrow$ # of iterations

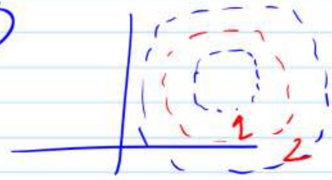
E-step:- $O(mK \cdot n)$ M-Step $= O(mn)$

Total Time $O(TmKn)$ ↳ $\sum_k m_k n = m\overline{n}$ ⇓

(A)



(B)



↳ (A) Clusters are of (~) "equal" size
(B) They are "well" separated

GMM :- Gaussian Mixture Models

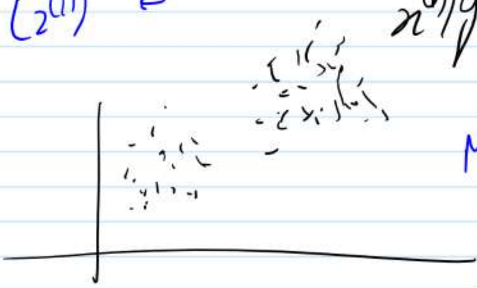
GDA ⇓

$\{x^{(i)}, \cancel{y^{(i)}}\}_{i=1}^{m}$ $y^{(i)} \sim$ Bernoulli(ϕ) Multinoulli(Φ)

Hidden ($z^{(i)}$)

$x^{(i)} | y^{(i)} = k \sim N(\mu_k, \Sigma_k)$

hidden class variable



M-step (1) if we knew $y^{(i)}$'s $z^{(i)}$'s
⇒ Find $(\mu_k, \Sigma_k)\ \forall k$ (Φ)

E-step (2) if we knew θ, then
estimate $P(z^{(i)} | x^{(i)}; \theta)$
$\propto P(x^{(i)} | z^{(i)}; \theta) P(z^{(i)}; \theta)$